Name
Date
Course

# The Sum and Difference Identities (Homework)

Find the values of the following expressions without a calculator.

1. $\sin 165^\circ$

2. $\cos 345^\circ$

3. $\sin 285^\circ$

4. $\cos 195^\circ$

⑤ $\sin 15^\circ$

Simplify the following expressions. Write exact expressions.
In other words, you do not have to use a calculator.

⑥ $\sin 25A \cos 15A + \sin 15A \cos 25A$

⑦ $\cos\alpha \cos 3\alpha + \sin\alpha \sin 3\alpha$

⑧ $\sin 30^\circ \cos 20^\circ - \sin 20^\circ \cos 30^\circ$

⑨ $\cos 56^\circ \cos 2^\circ - \sin 56^\circ \sin 2^\circ$

Prove that the following equations are true by converting the left side to the right.

⑩ $\sin(\theta - W) + \sin(\theta + W) = 2\sin\theta \cos W$

⑪ $\cos 6X - \cos 14X \cos 8X = \sin 14X \sin 8X$

⑫ $\cos 72^\circ = \cos 10^\circ \cos 62^\circ - \sin 10^\circ \sin 62^\circ$

(13) $\sin(A+90^\circ) = \cos A$

(14) $1 - \cos(B - 180^\circ) = \cos B + 1$

Use class problem #29 to help you find the value of the expression below without a calculator. Hint: Use a ratio identity.

(15) $\cot 255^\circ$

# Homework Answers

1. $\boxed{\dfrac{\sqrt{6}-\sqrt{2}}{4}}$

2. $\boxed{\dfrac{\sqrt{6}+\sqrt{2}}{4}}$

3. $\boxed{-\dfrac{\sqrt{6}+\sqrt{2}}{4}}$

4. $\boxed{-\dfrac{\sqrt{6}+\sqrt{2}}{4}}$

5. $\boxed{\dfrac{\sqrt{6}-\sqrt{2}}{4}}$

6. $\boxed{\sin(40A)}$

7. $\boxed{\cos(-2\alpha)}$

8. $\boxed{\sin 10^\circ}$

9. $\boxed{\cos 58^\circ}$

10. $\sin(\theta - W) + \sin(\theta + W) = 2\sin\theta\cos W$

$\sin\theta\cos W - \sin W\cos\theta + \sin\theta\cos W + \sin W\cos\theta =$

$2\sin\theta\cos W =$

11. $\cos 6x - \cos 14x\cos 8x = \sin 14x\sin 8x$

$\cos(14x - 8x) - \cos 14x\cos 8x =$

$\cos 14x\cos 8x + \sin 14x\sin 8x - \cos 14x\cos 8x =$

$\sin 14x\sin 8x =$

12. $\cos 72^\circ = \cos 10^\circ\cos 62^\circ - \sin 10^\circ\sin 62^\circ$

$\cos(10^\circ + 62^\circ) =$

$\cos 10^\circ\cos 62^\circ - \sin 10^\circ\sin 62^\circ =$

13. $\sin(A + 90^\circ) = \cos A$

$\sin A\cos 90^\circ + \sin 90^\circ\cos A =$

$(\sin A)(0) + (1)\cos A =$

$\cos A =$

14. $1 - \cos(B - 180^\circ) = \cos B + 1$

$1 - [\cos B\cos 180^\circ + \sin B\sin 180^\circ] =$

$1 - [(\cos B)(-1) + (\sin B)(0)] =$

$1 - [-\cos B + 0] =$

$1 + \cos B =$

$\cos B + 1 =$

15. $\boxed{2-\sqrt{3}}$